

Rahim Iqbal

[in Rahim-Iqbal](#) | [✉ rahimi.23@cse.mrt.ac.lk](mailto:rahimi.23@cse.mrt.ac.lk) | [☎ +94-75-9257307](tel:+94-75-9257307) | [🔗 Rahim-Iqbal](#) | [📁 Portfolio](#) | [📍 Gampaha](#)

Computer Science and Engineering undergraduate at the University of Moratuwa, specializing in Data Science. Experienced in applied machine learning, NLP, and building full-stack systems that hold up outside a demo. Seeking an ML Engineering internship to contribute production skills in PyTorch, FastAPI, Docker, and Kubernetes to CML Insight's enterprise AI infrastructure.

EDUCATION

UNIVERSITY OF MORATUWA

2024 - Present

B.Sc. (Hons) in Engineering, Specialized in Computer Science Engineering
Stream: Data Science (3rd year)

ZAHIRA COLLEGE COLOMBO

2009 - 2023

General Certification of Education Advanced Level - Physical Science Stream
Distinctions for Combined Mathematics, Physics, Chemistry

RESEARCH AND SELF PROJECTS

Environmental Drivers of Respiratory Disease: A District Level Analysis [🔗](#)

Jan 2026 - May 2026

Best Paper Award Nominee ·

Accepted, MERCon 2026, Moratuwa Engineering Research Conference [arXiv:2607.04416](#) [Dataset](#)

- Assembled an 11-year panel dataset across all 25 Sri Lankan districts from satellite, pollutant and health admission sources
- Built temporally validated XGBoost models for respiratory admission rate ($R^2 = 0.937$) and monthly PM2.5 ($R^2 = 0.976$); SHAP identified air quality burden as the dominant driver (80.1%)

Audiolit [🔗](#)

Interactive multimodal Explainable-AI workbench for speech recognition, GitHub emotion analytics, and deepfake detection

Jul 2026 - Oct 2026

- Built an XAI platform extending the ECHO 1.0 baseline to concurrently run ASR, speech emotion recognition, and audio deepfake detection, with acoustic profiling and accent-bias auditing
- Implemented an asynchronous Celery/RQ task fabric for non-blocking inference, a deterministic SHA-256 content-addressed cache, and visual XAI methods like Grad-CAM and Integrated Gradients
- Shipped the React web client with canvas-driven spectrogram mutation, real-time WebSocket progress tracking, and interactive 2D/3D latent-space projections of hidden states
- Tools: React, TypeScript, FastAPI, PyTorch, Hugging Face, Captum, Celery, Redis, MongoDB, Librosa, Docker

DiffDetect-YOLO [🔗](#)

April 2025 - May 2025

- Developed a computer vision pipeline to automate the detection of visual differences between paired images for a Kaggle competition.
- Leveraged YOLOv8-World for open-vocabulary zero-shot detection, successfully localizing anomalies and bounding box generation for altered image regions.

GoldNest [🔗](#) [Live](#)

March 2025

Micro-investment fintech platform, web and mobile

- Built a platform for micro-investing in fractional gold from Rs. 100, with wallets, automated savings plans and physical redemption
- Tokenised holdings on Polygon for verifiable ownership, and added AI-driven market analysis and a gamification layer
- Shipped the companion iOS and Android client with biometric login, live pricing charts and instant buy and sell
- Won 1st place at Dev{thon} 2.0 and reached the Startup Spark 2.0 finals, with team Code Mavericks
- Tools: Next.js, React, Node.js, Express, MongoDB, OpenZeppelin, Polygon, Groq SDK, Chart.js, JWT.

MediBridge [🔗](#)

February 2025 - April 2025

Interoperable digital health passport

- Engineered the backend entirely in Ballerina so patients consolidate records across institutions and grant providers scoped access
- Split records, consent and assist into independent services behind a central gateway, with role-based access control

• Tools: Ballerina, REST APIs, Docker

SKILLS

TECHNICAL SKILLS:	Machine Learning	: PyTorch, scikit-learn, XGBoost, SHAP, Pandas, NumPy, Jupyter, Yolo
	Frontend	: React, Next.js, Tailwind CSS, Figma
	Backend	: FastAPI, Flask, REST APIs, Microservices, Apache Kafka
	Database	: PostgreSQL,MySQL,
	DevOps	: Docker, Kubernetes, GitHub Actions, AWS
	Programming Languages	: Python, TypeScript, JavaScript, SQL, C++,Ballerina
	Tools	: Vscode, Pycharm, Clion, Chrome DevTools, Huggingface

LANGUAGES: English, Sinhala, Tamil

ACHIEVEMENTS & AWARDS

1ST PLACE, DEV{THON} 2.0 2025	Built GoldNest with team Code Mavericks, recognized for innovation in the final round. JISC eduCLaaS & Rotaract Club, University of Moratuwa
2ND RUNNER-UP, OCTWAVE 2.0 2025	Placed third among the top AI/ML teams from Sri Lankan universities, across machine learning and computer vision. Competition partnered with Tortil Inc. IEEE IAS Student Branch Chapter, University of Moratuwa
FINALIST, STARTUP SPARK 2.0, TECHNO 2025	Pitched GoldNest with team Code Mavericks at the “Innovate for Impact” startup competition, held at the BMICH, Colombo. IESL Student Chapter, University of Moratuwa.
FINALIST, AITHON 2026	Advanced to the MVP Development Phase with team MatriXplorers, out of 60+ teams. Enterprise AI Innovation Challenge, Hemas Holdings PLC May 2026

EXTRA CURRICULAR ACTIVITIES

YOUTH INITIATIVE CIRCLE	Active volunteer and mentor in a grassroots community initiative dedicated to supporting my home village. Facilitate educational guidance and skill-building activities to empower local youth and foster community development.
MAJLIS UL ISLAM	Assisted Majlis ul Islam with first-year student mentorship, Ramadan logistics, and social media outreach campaigns.

CERTIFICATIONS

MICROSERVICES AND CI/CD PIPELINE BUILDER	AWS ACADEMY (JAN 2026)
CLOUD FOUNDATIONS	AWS ACADEMY (JAN 2026)
GENERATIVE AI MASTERMIND	OUTSKILLS (2026)

REFEREES

PROF. UTHAYASANKER THAYASIVAM Former Head of the Department, Senior Lecturer Department of Computer Science and Engineering University of Moratuwa, Sri Lanka. <i>rtuthaya@cse.mrt.ac.lk</i>	PROF. DULANI MEEDENIYA Senior Lecturer Department of Computer Science and Engineering University of Moratuwa, Sri Lanka. <i>dulanim@cse.mrt.ac.lk</i>
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